Beyond Outcome Disparities: A Sensitivity Framework for Auditing Algorithm Fairness

(And Application to Pandemic Modeling)

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Workshop: Models of Pandemics - February 3, 2026

The Algorithmic Fairness Problem

Algorithms increasingly govern high-stakes decisions, but traditional fairness metrics are insufficient

Traditional Approach

Demographic parity, equalized odds, calibration
Focus on outcome disparities across groups

What's Missing

Why do disparities occur? Which features drive them?
Are decision rules stable over time or context?

Our Approach

Apply Global Sensitivity Analysis to fairness auditing:
diagnose transmission mechanisms of bias

Optimal Transport-based Global Sensitivity Analysis

Quantifies how model outputs respond to input feature variations using Wasserstein distance.
Measures the 'effort' required to transform output distribution when feature values change.

Why OT-GSA?

Distribution-preserving:
Respects full output distributions

Non-parametric:
No assumptions about model structure

Interpretable:
Sensitivity indices directly measure feature importance

Key Innovation

Beyond traditional fairness auditing:

Compute sensitivity indices separately for different groups, revealing:

→ Which features drive disparities
→ Whether features affect groups asymmetrically
→ How importance changes over time or context

Four-Dimensional Taxonomy

Each dimension captures a distinct fairness principle

1. Magnitude (S_j)

Proportionality Principle
Models shouldn't over-rely on any single feature. High S_j indicates disproportionate influence.

2. Asymmetry (ΔS_j)

Equal Treatment Principle
Different groups should face equivalent decision rules. $\Delta S_j > 0.15$ signals asymmetric treatment.

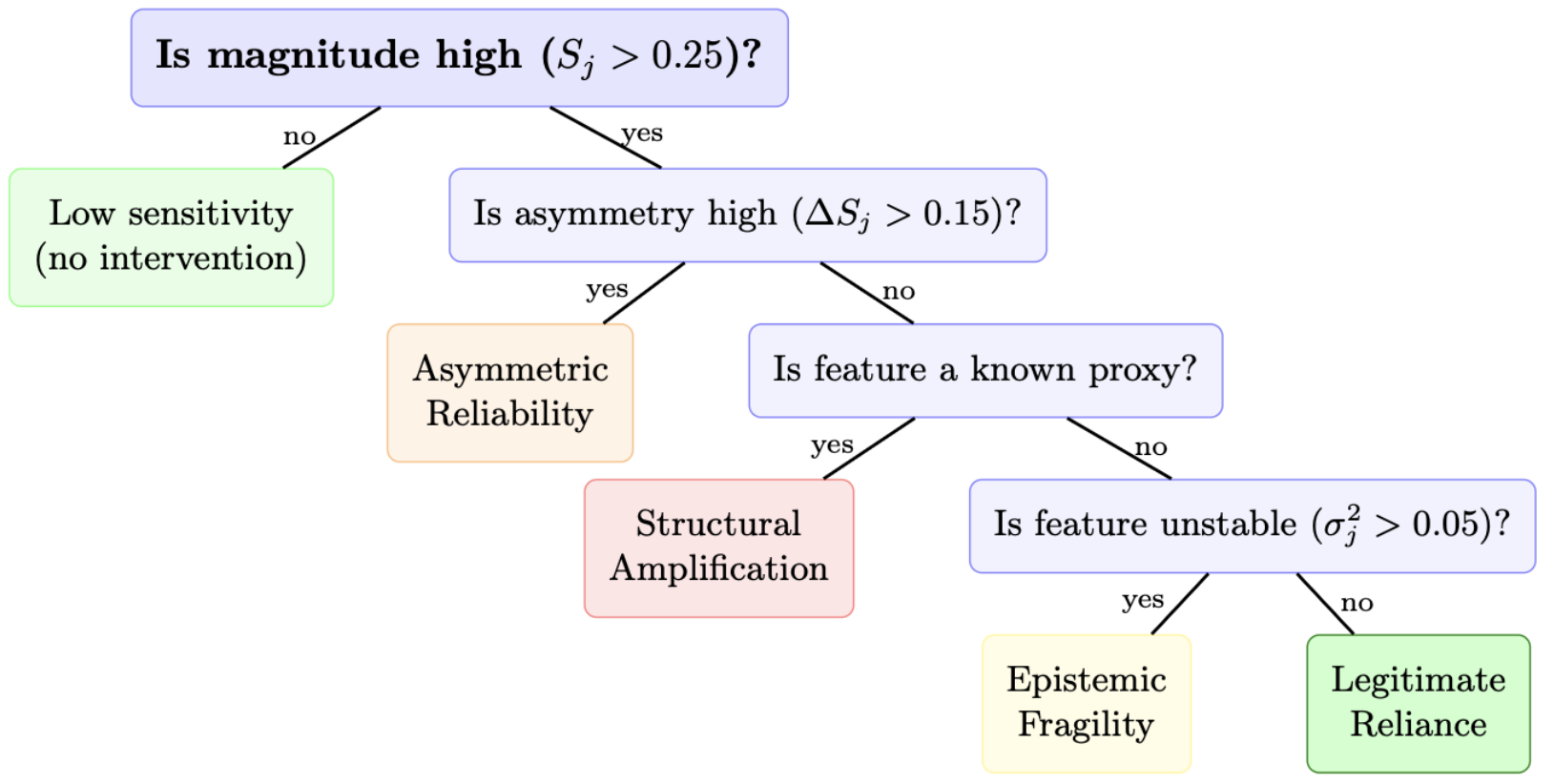
3. Justification ($X_j \in P$)

Non-Discrimination Principle
Predictions shouldn't depend on proxies (P) for protected attributes (race, gender, socioeconomic status).

4. Stability (σ_j^2)

Epistemic Robustness
Decisions shouldn't depend on temporally fragile measurements. High σ_j^2 indicates instability.

Four-Dimensional Taxonomy



Case Study 1: COMPAS Background

Correctional Offender Management Profiling for Alternative Sanctions

Commercial algorithm used by courts to predict recidivism risk (likelihood of re-offense)
Widely criticized after ProPublica investigation revealed racial bias in predictions

Dataset

7,214 defendants from Broward County, Florida (2013-2014)

Features: age, priors_count, charge degree, gender, race

Outcome: Two-year recidivism

Research Question

Do features exhibit asymmetric importance across protected groups (age, race)?

Can we identify which features are proxies for discrimination vs. legitimate predictors?

COMPAS: Results from Global Sensitivity Analysis

	African-American		Caucasian		Asymmetry
Feature	FPR	FNR	FPR	FNR	ΔS_j
priors_count	0.25	**0.34**	0.18	0.27	**0.16**
age	0.20	0.15	**0.40**	0.31	**0.25**
juv_fel_count	<0.01	0.03	<0.01	<0.01	0.03
juv_misd_count	0.02	0.05	<0.01	0.03	0.03
juv_other_count	<0.01	0.03	0.04	0.03	0.04
c_days_from_compas	0.04	0.02	0.02	0.03	0.02

Feature	**Error**	**Max S_j**	ΔS_j	**Pattern**
age	FPR	0.40	**0.25**	Asymmetric Reliability
age	FNR	0.31	0.16	Asymmetric Reliability
priors_count	FNR	0.34	0.07	Structural Amplification
priors_count	FPR	0.25	0.07	Structural Amplification
juv_fel_count	FNR	0.03	0.03	Low Sensitivity
juv_fel_count	FPR	<0.01	<0.01	Low Sensitivity
juv_misd_count	FNR	0.05	0.03	Low Sensitivity
juv_misd_count	FPR	0.02	0.02	Low Sensitivity
juv_other_count	FPR	0.04	0.04	Low Sensitivity
juv_other_count	FNR	0.03	<0.01	Low Sensitivity
c_days_from_compas	FPR	0.04	0.02	Low Sensitivity
c_days_from_compas	FNR	0.03	<0.01	Low Sensitivity

COMPAS: Key Findings

Age Asymmetry ($\Delta S = 0.25$)

'age' has $S_j = 0.40$ for younger defendants but only $S_j = 0.15$ for older defendants
→ Violates equal treatment principle: different algorithmic treatment by age

❌ Feature Removal Fails

Removing biased features doesn't fix the problem—it redistributes sensitivity to correlated proxies:

Before: prior counts ($S_j = 0.34$)

After removal:
age + charge_degree collectively
$S_j = 0.52$

Problem persists in different form!

✓ Insight: System Redesign

GSA reveals that piecemeal feature removal is insufficient.

Solution requires:

- Holistic system redesign
- Address correlation structure
- Stratified models for different age groups ($\Delta S > 0.15$)
- Continuous asymmetry monitoring

Case Study 2: CMS Hospital Readmission Background

Hospital Readmissions Reduction Program (HRRP)

Medicare program that financially penalizes hospitals with higher-than-expected 30-day readmission rates. Uses algorithmic risk adjustment to compare hospital performance.
Policy transition: 2016 Cures Act changed data collection methods

Dataset

4,806 hospitals across three eras:

- Pre-Cures (FY 2016-2017)
- Post-Cures (FY 2018-2019)
- COVID-Recovery (FY 2020+)

Features: Prior_ERR, Region, Volume_Log, High_Poverty_State

Research Question

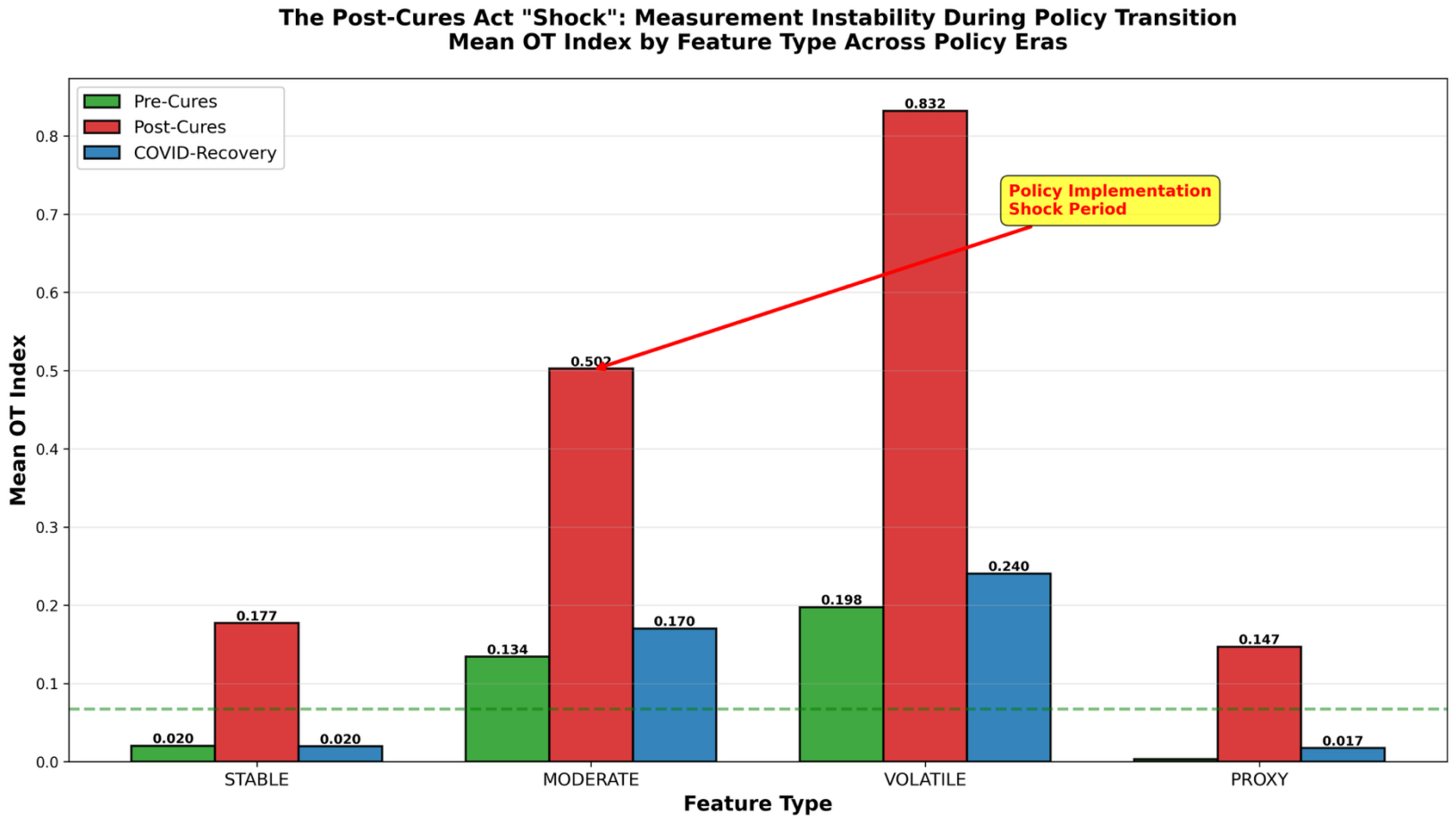
Do policy transitions create measurement system shocks?

Can we detect temporal instability in feature importance (σ_j^2) that creates epistemic unfairness?

CMS: Results - Measurement System Shock

Rank	Feature	Mean S_j	σ_j^2	Classification
High Importance + High Fragility (Danger Zone):				
1	`Prior_ERR`	0.42	**0.10**	Epistemically Fragile
2	`Region`	0.17	**0.07**	Epistemically Fragile
3	`Volume_Log`	0.44	**0.06**	Epistemically Fragile
Moderate Importance + Moderate Stability:				
4	`Ownership_Numeric`	0.21	0.05	Epistemically Fragile
5	`Quality_Rating`	0.30	0.04	Moderately Unstable
6	`Uses_EHR`	0.07	0.03	Moderately Unstable
Low Importance + Perfect Stability:				
7	`Has_Emergency`	<0.01	<0.01	Stable
8	`Hospital_Type_Numeric`	<0.01	<0.01	Stable
9	`High_Poverty_State`	<0.01	<0.01	Stable
10	`Medicaid_Expansion_State`	<0.01	<0.01	Stable

CMS: Results - Measurement System Shock



CMS: Key Findings

Universal Amplification During Policy Transition

Post-Cures era: **All** features simultaneously inflated (+270% to +4,200%)—not redistribution but universal amplification. Signals systemic measurement failure, creating epistemic unfairness: identical performance evaluated differently depending solely on timing

Epistemic Fragility

Temporal variance:

- Prior_ERR: $\sigma^2 = 0.10$
- Region: $\sigma^2 = 0.07$
- Volume_Log: $\sigma^2 = 0.06$

All exceed threshold $\sigma^2 > 0.05$, indicating temporally fragile measurements unsuitable for high-stakes decisions

Technical Fix Backfires

Attempted solution: Smooth Prior_ERR with rolling average

Result: Instability increased 13% (paradoxical!)

Feedback loops require policy redesign, not engineering patches

Epistemic Unfairness

Identical hospital performance receives radically different penalties depending solely on timing of evaluation

Violates procedural fairness: evaluation systems become fundamentally unpredictable

Traditional metrics miss this!

Key Insights from Global Sensitivity Analysis

1. Fairness is Multi-Dimensional

Traditional metrics (demographic parity, equalized odds) are necessary but insufficient. Epistemic robustness and procedural transparency are equally critical.

2. System-Level Interventions Required

Feature removal redistributes sensitivity. Technical smoothing can paradoxically increase instability. Holistic system redesign with continuous monitoring is necessary.

3. Measurement System Shocks Create New Unfairness

Policy transitions can destabilize all features simultaneously, creating epistemic unfairness invisible to traditional cross-sectional audits. Longitudinal GSA is essential.

4. Prospective Analysis Essential

Fairness auditing must evolve from post-hoc validation to prospective sensitivity analysis—anticipating how changes propagate through sociotechnical systems.

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Application to Pandemic Modeling

Can Global Sensitivity Analysis help evaluate models when pandemic features are poorly known?

Why pandemics are a fairness problem

Pandemic models as decision systems

- Pandemic models do not only predict cases or deaths
- They justify coercive public decisions:
  - Lockdowns and mobility restrictions
  - ICU triage and hospital access
  - Vaccine prioritization

Core claim

- Fairness failures can arise even with good intentions and accurate predictions
- The key issue is *how uncertainty is translated into decisions*

Limits of standard pandemic modelling

What standard sensitivity analysis focuses on

- Output uncertainty (cases, deaths, hospitalisations)
- Parameter uncertainty (R_0, IFR, contact rates)

What it misses

- Distributional effects across populations
- Stability of decision rules over time
- Reliance on proxies under emergency conditions

Result

- Models may be epidemiologically sound but normatively fragile

GSA for Pandemic Models: The Problems

Temporal Instability: SIR Model Parameters

R_0 estimates varied wildly: 1.4 to 6.5 across studies | Contact rates changed with policy
Model predictions diverged dramatically based on timing | Variant emergence = shocks

Result: Identical interventions evaluated differently depending on when models ran ($\sigma^2 >> 0.05$)

Geographic Asymmetry & Proxy Variables

ICU capacity models relied more on demographics in urban vs. rural areas ($\Delta S > 0.15$)
ZIP code became high-importance for mortality risk—but proxies for race, SES, healthcare access

Result: Same performance, different evaluations based on location—violates equal treatment

GSA for Next Pandemic: What Should We Do?

1. Pre-Crisis Protocol Development

Establish GSA frameworks **before** the next pandemic with pre-specified thresholds, monitoring frequency, and intervention triggers

2. Continuous Temporal Monitoring

Deploy automated Σ^2system tracking with >50% increase alerts triggering automatic review and potential enforcement suspension until stability restores

3. Stratified Model Development

Build separate models for populations with $\Delta S > 0.15$ (urban/rural, age groups) rather than forcing universal algorithms that treat dissimilar contexts identically

4. Proxy Feature Registry

Maintain updated catalogs of known proxies (P) for rapid Tier 1 screening during emergency model deployment (e.g., ZIP code as race/SES proxy)

5. Interdisciplinary Response Teams

Integrate ML researchers, epidemiologists, ethicists, and legal scholars **before** crisis deployment—not as *post-hoc* correction

Conclusions

- Algorithmic fairness is not a property to be measured but a multi-dimensional system quality to be audited, monitored, and maintained.

- For pandemics: the question is not just 'are decisions fair?' but 'are decision systems knowable?' When evaluation systems become fundamentally unpredictable, even calibrated algorithms violate fairness principles.

- Global Sensitivity Analysis provides the technical foundation to make algorithmic systems epistemically robust, procedurally transparent, and temporally stable—properties essential for justice in an age of automated decision-making.

Thank You

Questions & Discussion

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February 3, 2026 | BSM-UPF Barcelona